YUZHE ZHOU

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EDUCATION:

Purdue University West Lafayette, IN, US
Doctor of Philosophy - PhD, Computer and Information Technology Expected: May 2027
Committee: *Dr. Tianyi Li (Chair), Dr. Alejandra J. Magana, Dr. Ming Yin, Dr. Paul J. Thomas*

Purdue University West Lafayette, IN, US
Master of Science - MS, Computer and Information Technology August 2022 - May 2024
Thesis: *Towards Contextualized Programming Education by Developing a Learnersourcing Workflow*
Advisor: *Dr. Tianyi Li*

Purdue University West Lafayette, IN, US
Bachelor of Science - BS, Computer and Information Technology August 2018 - May 2022

SKILLS:

Programming Languages: SQL, PL/SQL, Python, C#, R, SPSS, JavaScript, HTML, CSS,
Data & Analytics Tools: Snowflake, Tableau, Power BI, Excel (Advanced)
Software & Platforms: Cursor (AI-powered IDE), Jupyter Notebook, Qualtrics, MTurk
Research Skills: Research Design, Literature Review, Qualitative Data Analysis, Quantitative Data Analysis, Inter-Rater Reliability, Mixed-Method
Languages: Chinese Mandarin (Native), English (Proficient)

RESEARCH EXPERIENCE:

Purdue University West Lafayette, IN
Graduate Research Assistant August 2022 – Present

Project: ExPeerience – An AI-Assisted Learnersourcing System for Programming Learning **PI: Dr. Tianyi Li**

- Led the design and development of **ExPeerience**, a web-based application operationalizing the learnersourcing workflow that engaged students in co-creating contextualized and relatable learning materials in programming education.
- Collaborated with the PI and mentored student developers to implement the system using the **Django** framework, integrated with **PostgreSQL** for database management and **HTML, CSS, JavaScript, and Bootstrap** for the front end.
- Conducted **1:1 user studies with over 50 students** to evaluate usability, engagement, and impact on learning outcomes, and applied findings to guide iterative improvements.
- Leveraged **AI-assisted vibe coding tools (e.g., Cursor)** to accelerate iterative revisions and implemented **AI-powered features** to support students at different stages of the creation process - ideation, generation, and evaluation - enhancing usability, engagement, and scalability of the platform.
- Contributed to the research-to-practice transition by preparing the system for classroom deployment, supporting publications in top HCI and computing education venues.

Project: Learnersourcing Workflow Database Programming Education **PI: Dr. Tianyi Li, Dr. Alejandra J. Magana**

- Conducted an in-depth analysis of current learning methods and materials in programming education to assess their effectiveness and identify improvement opportunities.
- Developed and implemented a learnersourcing workflow that engaged over 270 students across five semesters in co-creating contextualized worked-out examples, improving learning transfer in database programming.
- Iteratively refined the workflow through multiple classroom deployments, applying evidence-based insights to optimize scalability and impact.
- Experimented with and integrated diverse educational tools and technologies to strengthen workflow adoption and classroom implementation, such as **MTurk, Qualtrics, and Circuits**.

Project: AI-Powered Grading and Feedback System for Computer Science Education **PI: Dr.Tianyi Li, Dr. Jian Zhao**

- Co-developed an **AI tool** with Purdue University and the University of Waterloo to provide personalized, standardized, and scalable feedback for coding assignments.
- Designed features such as a **smart comment bank** and error-detection capabilities (e.g., type mismatch, AI-generated code detection) to assist instructors and teaching assistants in grading large classes of 800–1,500 students.
- Conducted experiments with professors and teaching assistants, addressing challenges of **grading consistency, timeliness, and fairness** in computer science education.
- Advanced **human-AI collaboration research** ensured instructors remained in control of AI-assisted grading while enhancing efficiency and instructional quality.

Project: Context Familiarity on Student Learning in Computing Education **PI: Dr.Tianyi Li, Dr. Paul J. Thomas**

- Designed and executed a **mixed-methods quasi-experimental study** with 101 undergraduate students in an introductory database programming course.
- Investigated differences between instructor-defined and student-perceived context familiarity and their impact on students' SQL performance and confidence.
- Applied **statistical analyses** (Mann-Whitney U tests, paired t-tests, regression models) to examine the relationship between **context familiarity, initial confidence, and learning outcomes**.
- Found that while both familiar and unfamiliar contexts improved learning, familiar contexts significantly boosted student confidence, highlighting the role of perceived familiarity in motivation and engagement.

Purdue University West Lafayette, IN
Undergraduate Research (with Dr. Alejandra J. Magana) September 2021 – December 2021

- Conducted a focused literature review and defined research questions to frame the study's objectives.
- Performed both qualitative and quantitative data analysis, ensuring rigor through systematic inter-rater reliability.
- Independently analyzed a designated portion of the dataset, contributing significantly to the overall research findings.
- Authored a detailed technical report, which served as the foundation for a peer-reviewed journal article as a co-author.

RESEARCH INTERESTS:

Human-Computer Interaction, Human-AI Interaction, AI in Education, Computer Science Education

PUBLICATIONS:

Zhou, Y., Babu, P. M., Pandey, U., J. Magana, A., & Li, T. (2026, March). ExPeerience: Towards AI-Assisted Learnersourcing to Bridge Conceptual Understanding and Problem Solving in Database Programming Education. In *Proceedings of the 31st International Conference on Intelligent User Interfaces* (pp. 1178-1201).

Zhou, Y., Manjunatha Babu, P., Pandey, U., Magana, A. J., & Li, T. (2026, March). ExPeerience: Support Student-AI Co-Creation in a Learnersourcing Workflow. In *Companion Proceedings of the 31st International Conference on Intelligent User Interfaces* (pp. 176-179).

Zhou, Y., Magana, A. J., & Li, T. (2026). *Farm harvests and airplane engines: Contextualizing database exercises in relatable contexts with a learnersourcing workflow*. In *Proceedings of the International Conference of Computer-Supported Collaborative Learning (CSCL2026).* **Accepted**.

Zhou, Y., Thomas, P., & Li, T. (2025, November). *Examining Perception Differences in Context Familiarity Between Students and Instructors in Computing Education: a Case Study on Introductory Database Concepts.* In *2025 IEEE Frontiers in Education Conference (FIE)* (pp. 1-8). IEEE.

Zhou, Y., Magana, A. J., & Li, T. (2025, February). Facilitating Student's Learning Transfer in a Database Programming Class. In *Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 1* (pp. 1330-1336).

Li, T., **Zhou, Y.**, & Magana, A. J. (2025, June). Towards Learnersourcing Relatable and Contextualized Learning Materials: An Exploratory Study in a Database Programming Class. In *Proceedings of the 30th ACM Conference on Innovation and Technology in Computer Science Education V. 1* (pp. 479-485).

Zhou, Y. (2024). *Towards Contextualized Programming Education by Developing a Learnersourcing Workflow* (Master's thesis, Purdue University).

Nanda, G., Jaiswal, A., Castellanos, H., **Zhou, Y.**, Choi, A., & Magana, A. J. (2023). Evaluating the coverage and depth of latent dirichlet allocation topic model in comparison with human coding of qualitative data: the case of education research. *Machine Learning and Knowledge Extraction*, *5*(2), 473-490.

TEACHING EXPERIENCE:

Purdue University West Lafayette, IN
Graduate Teaching Assistant - CNIT 37200 Database Programming
Spring and Fall 2022, Spring and Fall 2023, Fall 2024, Fall 2025, Fall 2026
Graduate Teaching Assistant - CNIT 32200 Research Methodology and Design
Fall 2026